

Instructions to Lab 1 Instructions on Simulation on Transmission Line Using CST Studio Suite

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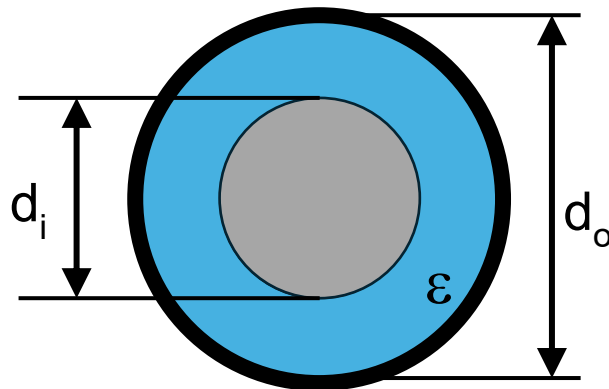
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Aim

To simulate parameters of a transmission line and check the matchiness to the load.

Background

The transmission line we are going to simulate is a coaxial cable, with characteristic impedance of $Z_0 = \frac{60}{\sqrt{\epsilon_r}} \ln \left(\frac{d_o}{d_i} \right)$, where ϵ_r is the relative permittivity of the insulator between the conductors, d_o is inner diameter of the outer conductor, d_i is the diameter of the inner conductor, as shown in the figure below.

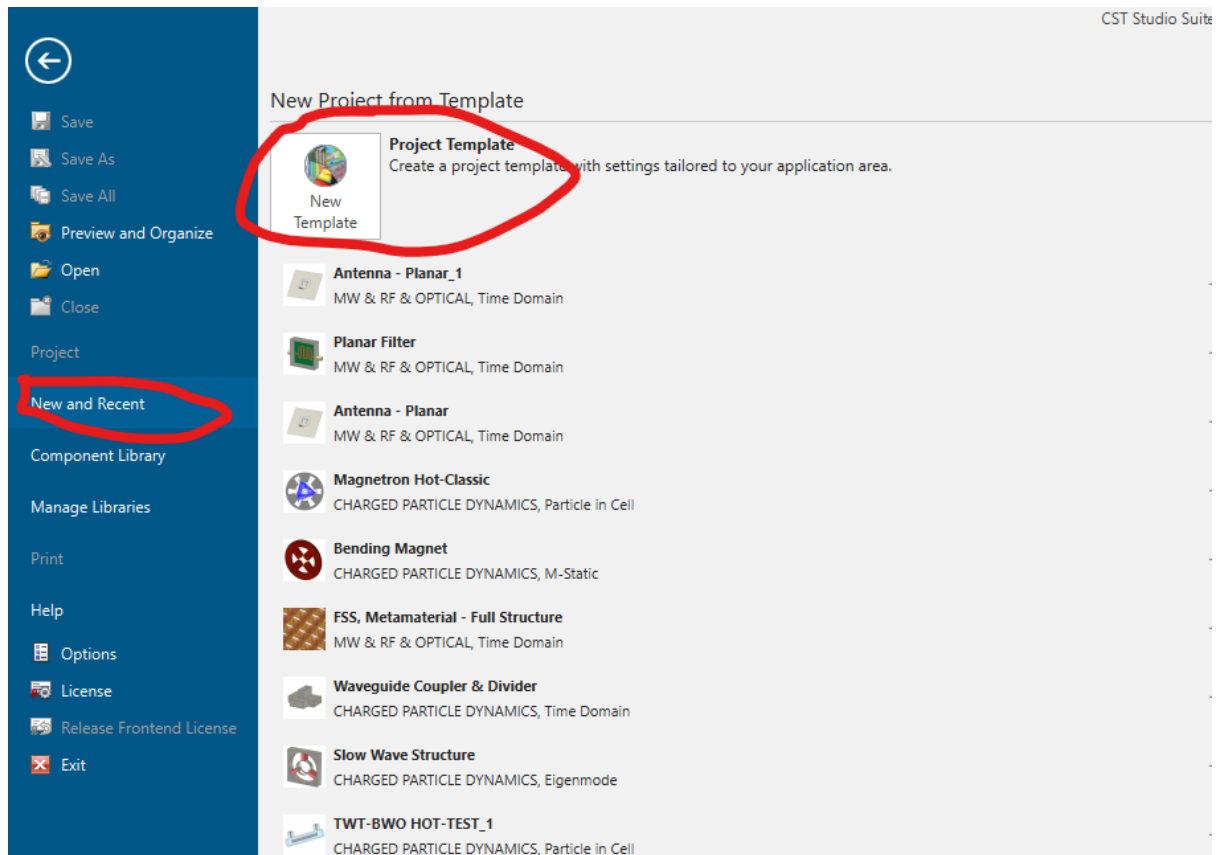


Procedures

The procedures of simulation in CST Studio Suite are as follows.

1. Open CST and Set a template.

Open CST Studio Suite, → New and Recent → New Template

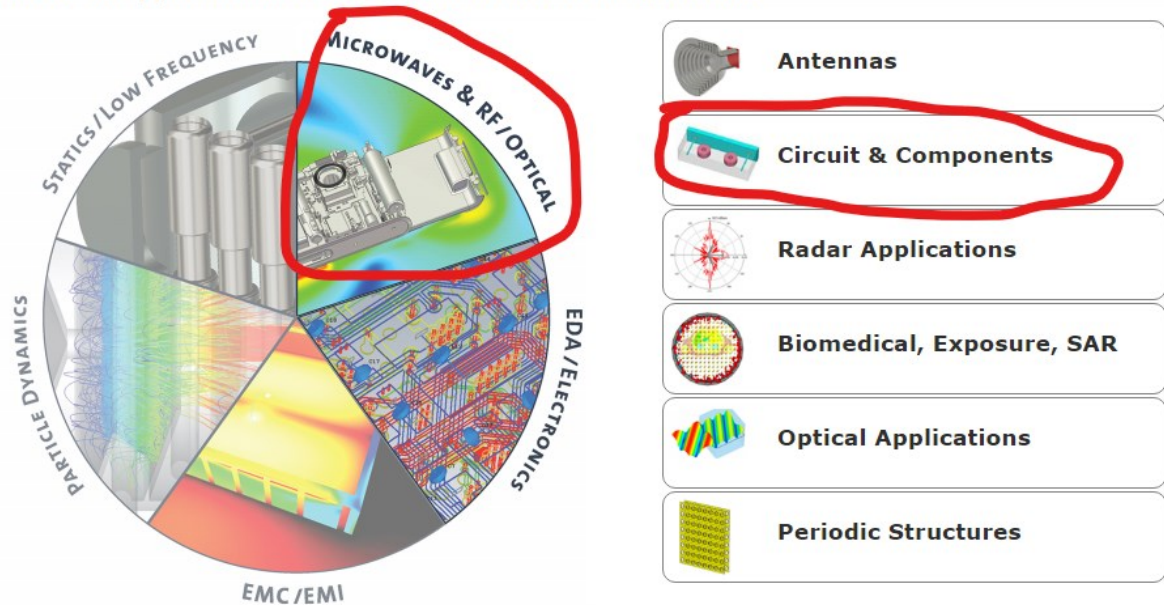


Then, choose Microwaves & RF/Optical → Circuit & Components

CST Studio Suite

Create Project Template

Choose an application area and then select one of the workflows:



Then choose Waveguide & Cavity Filters, then 'Next'

Create Project Template

MW & RF & OPTICAL | Circuit & Components

Please select a workflow:

Waveguide & Cavity Filters	Planar Filters	Waveguide Couplers & Dividers	Planar Couplers & Dividers
Coaxial (TEM) Connectors	Multipin - shielded Connectors	Multipin - unshielded Connectors	Packages

Then choose, 'Frequency Domain (fast reduced order)'

Create Project Template

MW & RF & OPTICAL | Circuit & Components | Waveguide & Cavity Filters | **Solvers** | Units | Settings | Summary

The recommended solvers for the selected workflow are:

 Frequency Domain (Fast Reduced Order) <i>fast frequency sweep</i>	This one !!
 Frequency Domain (Fast Reduced Order - Hex.) <i>quick analysis, without losses</i>	
 Eigenmode <i>to study input or inter-resonator coupling</i>	
 Frequency Domain <i>general, most robust</i>	

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Then 'Next'

Create Project Template

MW & RF & OPTICAL | Circuit & Components | Waveguide & Cavity Filters | Solvers | Units | **Settings** | Summary

Please select the Settings

Frequency Min.: GHz

Frequency Max.: GHz

Monitors: ☐ E-field ☐ H-field ☐ Farfield ☐ Power flow ☐ Power loss

Define at: GHz
Use semicolon as a separator to specify multiple values.
e.g. 20;30;30.1;30.2;30.3

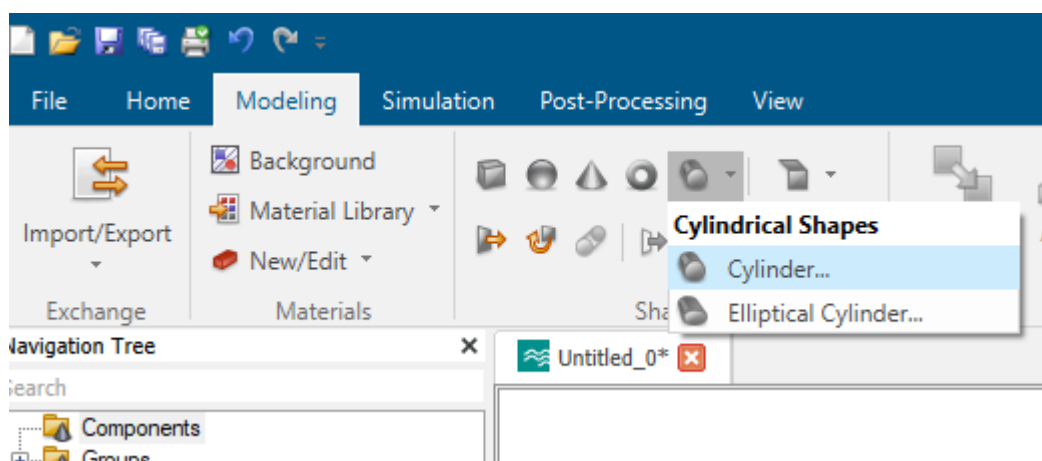
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Then set frequency range: 1-4 GHz. Then 'Finish'.

2. Building geometrical structures

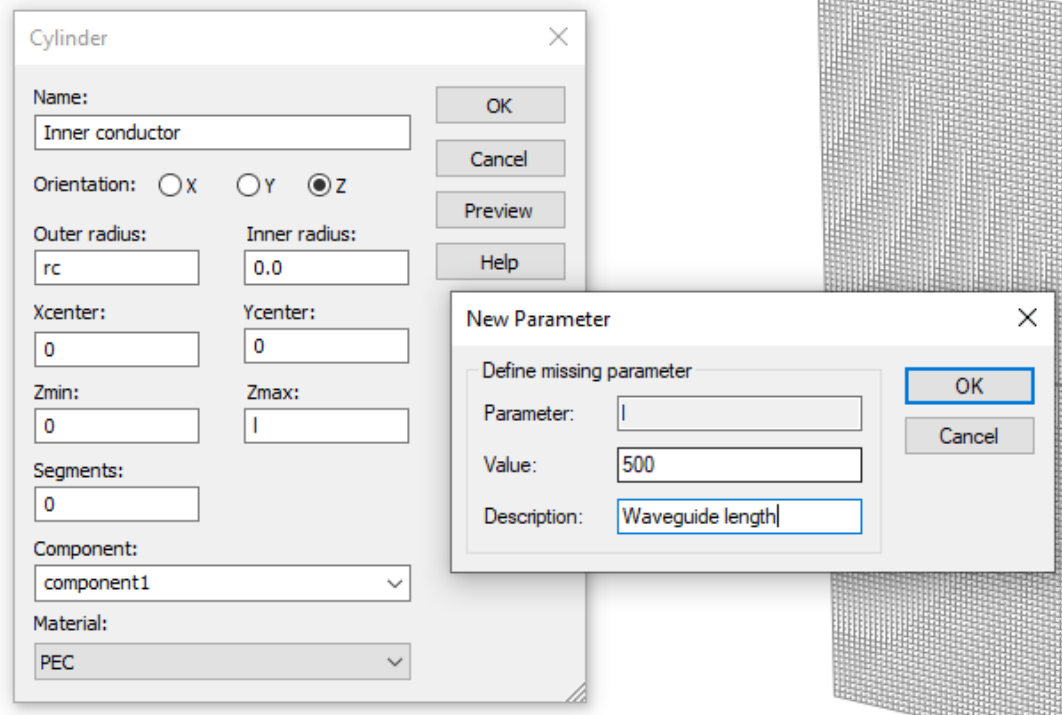
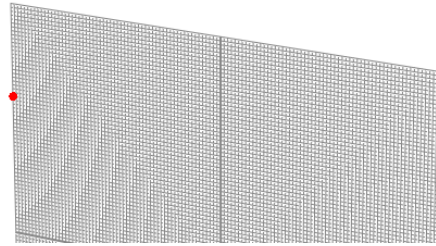
2.1 Build inner conductor.

Choose 'Modeling' → Cylindrical Shapes → Cylinder.



Then press ESC (on your keyboard!!!!) to show dialog box, as shown below.

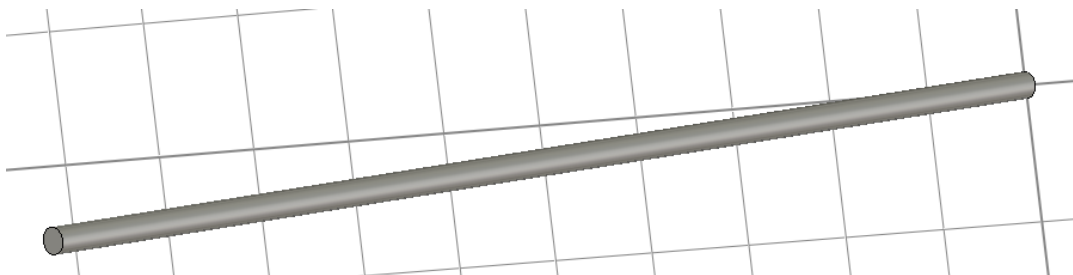
Double click center point in working plane (Press ESC to show dialog box)



Set the inner conductor parameters. You can either:

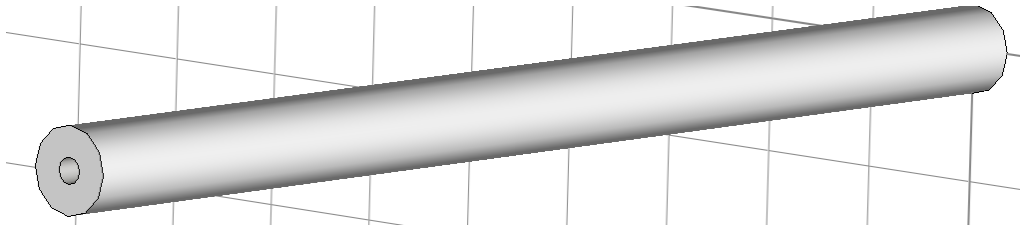
- Use parameters such as rc and l , for inner conductor radius and length. Set them as 5 mm and 500 nm;
- Or directly type in 5, and 500 as outer radius and Z_{max} .

You'll get a cylinder as follows. Using PEC for material.



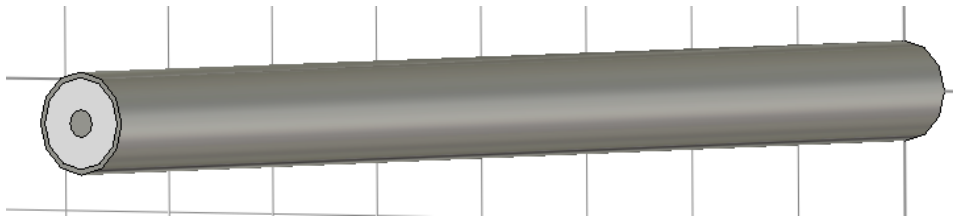
2.2 Building Teflon insulation

Similarly build insulation by creating cylinder with inner radius (r_c , the same as the previous step), outer radius ($r_t=33.5/2$), and the same length $l=500$ mm. Creating a new material for Teflon with $\epsilon_r=2.1$, and use this for the insulation.



2.3 Building Outer Conductor

Similarly build outer conductor by creating cylinder with inner radius ($r_t=33.5/2$), outer radius ($r_t=33.5/2+1$), and the same length $l=500$ mm, using PEC as material.

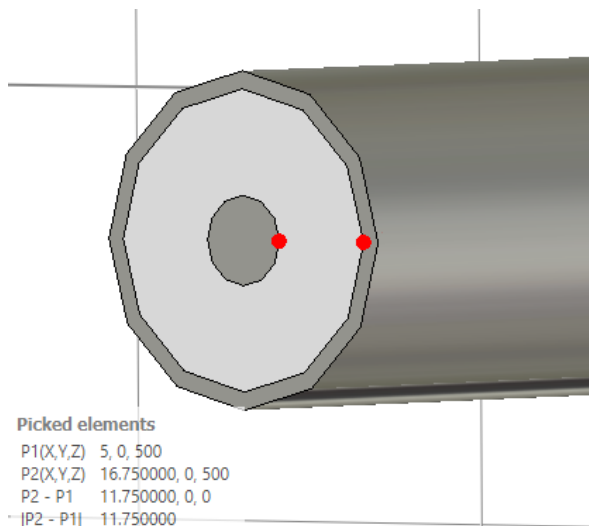


3. Setting 2 ports: port 1 and port 2 for source (generator) and load impedance.

Modelling → Pick point → Pick point on circle: Choose a point on outer edge of inner conductor.

Pick point → Pick point on circle: Choose a point on inner edge of outer conductor.

You have got 2 points.



Then Simulation → Discrete Port, Choose S-parameters, Set impedance as 50 Ohm. Click OK. Now you have defined a port.

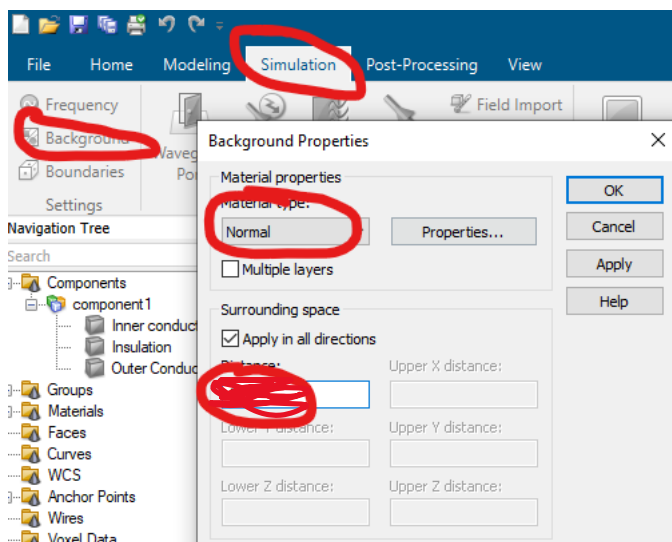
Rotate to the other end of the co-axial cable, define another port using the same way.

S parameters mean how much is transferred from one terminal to the other.

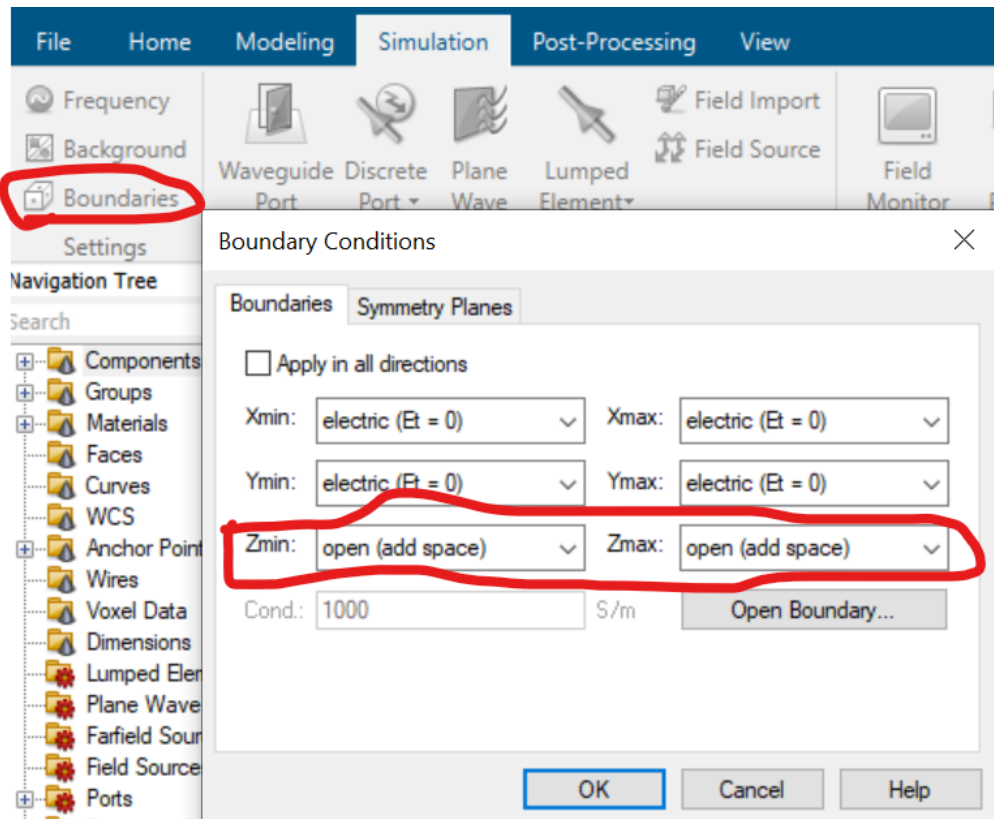
4. Simulation settings.

Check Simulation→Frequency, $f_{\min} = 1$ GHz, $f_{\max} = 4$ GHz.

Simulation→Background→Normal.

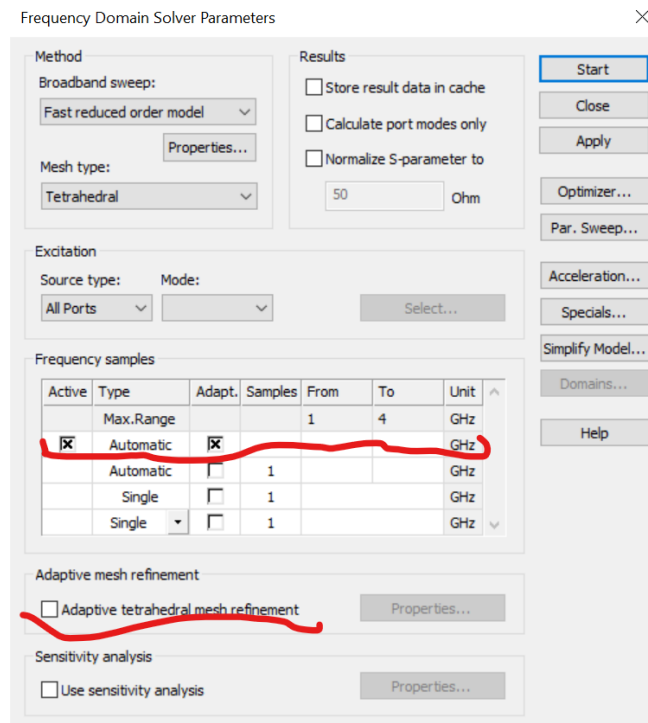


Boundaries→ Set $Z_{\min}$, $Z_{\max}$ as 'Open (add space)'.



5. Running simulation

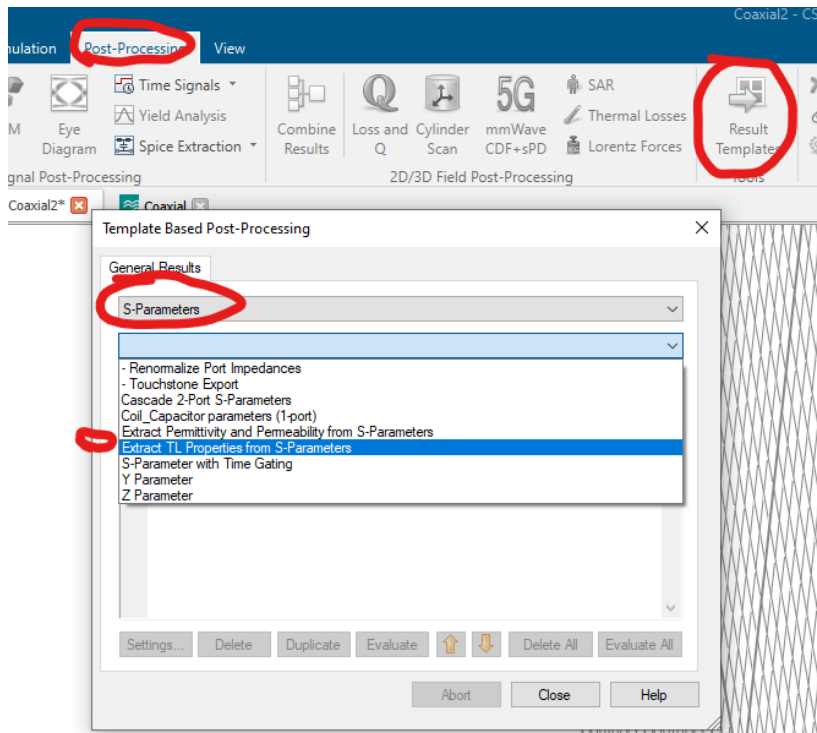
Simulation→F solver, and make the settings in the solver window.



Click on 'Start' to start simulation.

6. Post processing.

After simulation, set the post-processing: results templates → S parameters → Extract TL properties → Evaluate.



Check the results here.

